

Assignment 3

Bayesian Bernoulli Regression

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1 Posterior Formulation and Implementation

We analyze a binary-response dataset $\mathcal{D} = \{(y_i, x_i)\}_{i=1}^N$ with $N = 120$ observations, 54 successes, and covariate $x \in [-2.36, 2.44]$.

The main model (M1) is a Bayesian logistic regression:

$$y_i \mid \beta_0, \beta_1 \sim \text{Bernoulli}(\pi_i), \quad (1)$$

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 x_i, \quad (2)$$

with independent priors

$$\beta_0 \sim \mathcal{N}(0, 3^2), \quad \beta_1 \sim \mathcal{N}(0, 3^2). \quad (3)$$

The model was implemented in **JAGS** via the **rjags** interface in R. Before fitting, the covariate was standardised to zero mean and unit variance ($\tilde{x}_i = (x_i - \bar{x})/s_x$) to improve sampler efficiency and numerical stability. The JAGS model block specifies the Bernoulli likelihood with a logit link and encodes the prior precision as $1/9$ (equivalent to a standard deviation of 3):

```
model {  
  for (i in 1:N) {  
    y[i] ~ dbern(pi[i])  
    logit(pi[i]) <- beta0 + beta1 * x[i]  
  }  
  beta0 ~ dnorm(0, 1/9)  
  beta1 ~ dnorm(0, 1/9)  
}
```

2 MCMC Fitting and Diagnostics

Four independent chains are initialised from dispersed starting values $(\beta_0, \beta_1) \in \{-5, +5\}^2$, covering the extremes of the prior support. Each chain run for 1 000 burn-in iterations (discarded) followed by 20 000 sampling iterations, yielding 80 000 post-burn-in draws in total.

Figure 2 shows the trace plots for β_0 and β_1 . The dashed vertical line marks the end of the burn-in period. All four chains reach a common stationary regime well within the first 1 000 iterations, regardless of the dispersed starting values: the chosen burn-in is a conservative and sufficient value. After the burn-in, all chains overlap and fluctuate around a stable mean with no visible trend or drift.

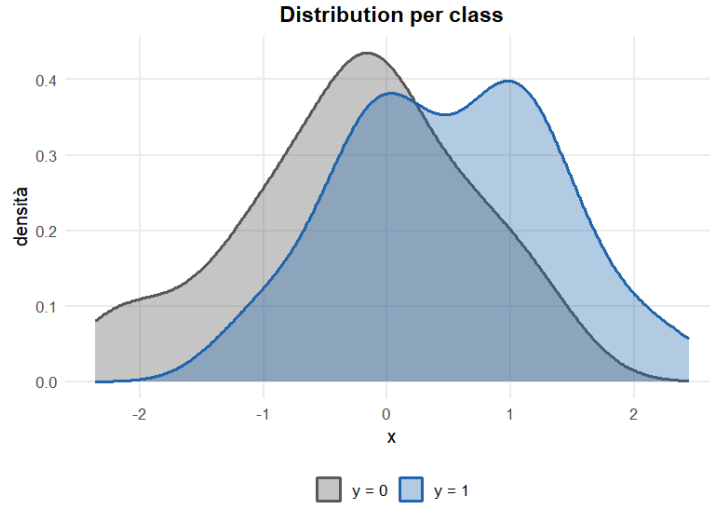


Figure 1: Distributions of the dataset

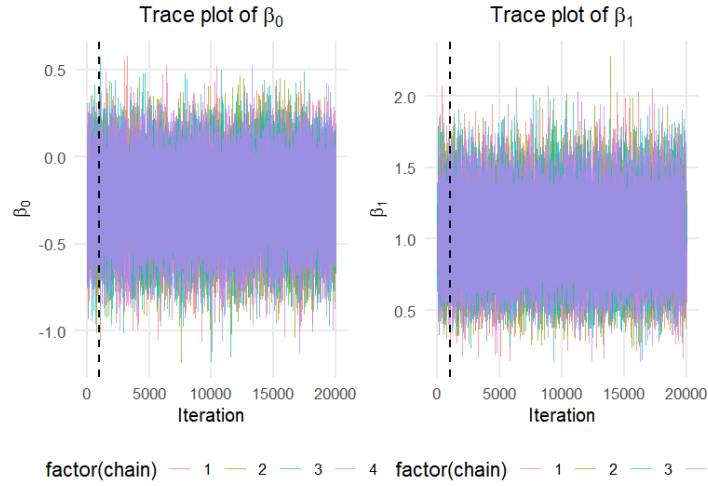


Figure 2: Trace plots for β_0 (left) and β_1 (right) across four chains. The dashed line at iteration 1 000 marks the end of the burn-in period.

The potential scale reduction factor is $\hat{R} = 1.00$ for both parameters (multivariate PSRF = 1), confirming that the chains have converged to the same stationary distribution.

The effective sample sizes are $\text{ESS}(\beta_0) = 47\,078$ and $\text{ESS}(\beta_1) = 48\,141$, representing approximately 59% efficiency relative to the nominal 80 000 draws. Both values are far above the commonly recommended threshold of 1 000, so posterior summaries based on these draws are highly reliable.

Table 1 reports the autocorrelation at selected lags. Autocorrelation decays to negligible levels by lag 5 ($|\text{ACF}| < 0.003$ for both parameters), confirming rapid mixing.

Table 1: Autocorrelation of the posterior draws at selected lags.

Lag	β_0	β_1
1	0.248	0.249
5	0.001	-0.002
10	-0.003	-0.001
50	0.010	0.010

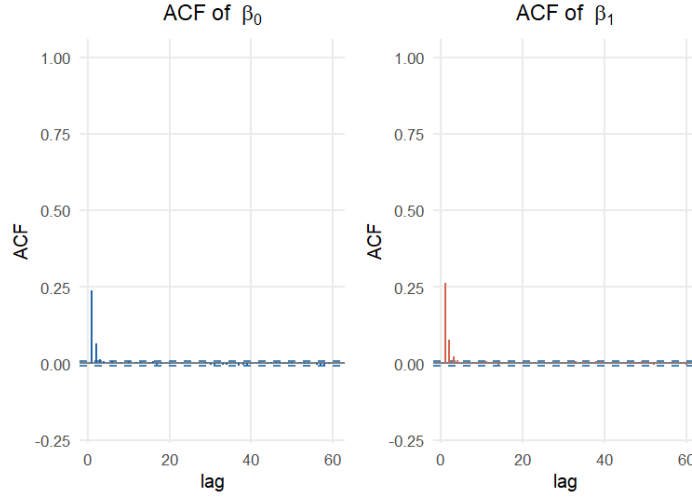


Figure 3: Autocorrelation function plots for β_0 (left) and β_1 (right). Dashed blue lines indicate the 95% significance band.

No thinning is applied. Since autocorrelation is negligible by lag 5 and the ESS already exceeds 47 000, thinning would only reduce the number of available samples without any gain.

3 Posterior Inference

Table 2 reports the posterior summaries for β_0 and β_1 . The marginal posterior distributions are shown in Figure 4, and the joint posterior samples in the (β_0, β_1) plane are displayed in Figure 5.

Table 2: Posterior summaries for the regression coefficients.

Parameter	Mean	Median	SD	2.5%	97.5%
β_0	-0.2562	-0.2553	0.2045	-0.6631	0.1405
β_1	1.0195	1.0111	0.2455	0.5657	1.5211

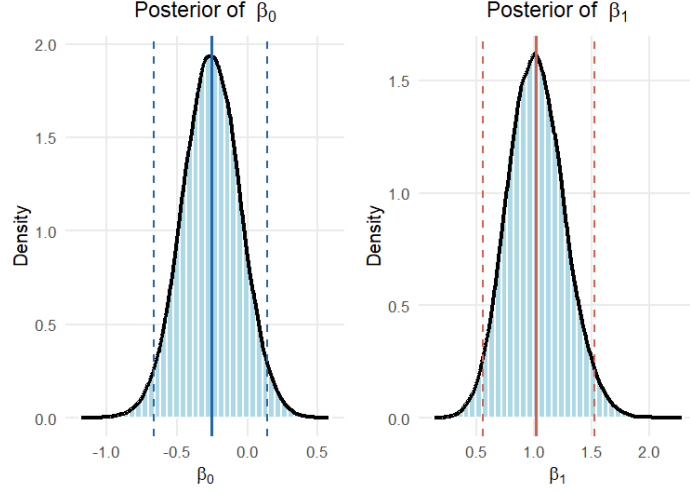


Figure 4: Marginal posterior distributions of β_0 (left) and β_1 (right). Solid vertical lines indicate the posterior mean; dashed lines indicate the 95% credible interval. The dotted line in the β_1 panel marks zero.

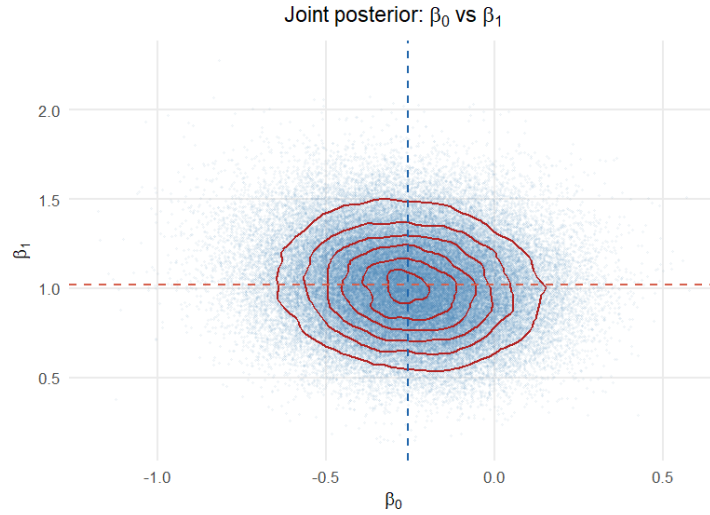


Figure 5: Joint posterior samples in the (β_0, β_1) plane with contour lines from a kernel density estimate. Dashed lines mark the posterior means.

The posterior mean of β_0 is -0.256 with a 95% credible interval of $[-0.663, 0.141]$. This reflects genuine uncertainty about the baseline log-odds: at the mean value of x , the probability of success is slightly below 0.5, but the data do not rule out $\beta_0 = 0$.

The posterior mean of β_1 is 1.020 and the 95% credible interval $[0.566, 1.521]$ lies entirely above zero, providing strong evidence of a positive association between x and the probability of success. A one-standard-deviation increase in x increases the log-odds of $y = 1$ by approximately 1.02, corresponding to an odds-ratio of $e^{1.02} \approx 2.77$. The posterior standard deviation of 0.25 is moderate relative to the effect size, confirming that the covariate effect is both practically and statistically meaningful.

The posterior correlation between β_0 and β_1 is -0.114 , which is weak. The near-orthogonality of the two parameters is a direct consequence of the standardisation of x . Centering the covariate at zero removes the linear dependence between intercept and slope that would otherwise arise, making the two parameters nearly independent a posteriori and facilitating efficient sampling.

4 Posterior Predictive Inference

Posterior predictive inference is carried out for a new observation at $\tilde{x}_{\text{new}} = 0$ on the standardised scale, corresponding to $x_{\text{new}} \approx 0.081$ on the original scale, i.e. approximately at the sample mean of x . For each posterior draw $(\beta_0^{(s)}, \beta_1^{(s)})$, we compute

$$\pi_{\text{new}}^{(s)} = \frac{\exp(\beta_0^{(s)} + \beta_1^{(s)} \tilde{x}_{\text{new}})}{1 + \exp(\beta_0^{(s)} + \beta_1^{(s)} \tilde{x}_{\text{new}})}, \quad (4)$$

and then draw $y_{\text{new}}^{(s)} \sim \text{Bernoulli}(\pi_{\text{new}}^{(s)})$. The results are summarised below:

$$\begin{aligned} \mathbb{E}[\pi_{\text{new}} \mid \mathcal{D}] &= 0.437, \\ \text{Median}(\pi_{\text{new}} \mid \mathcal{D}) &= 0.437, \\ \text{CI}_{95\%}(\pi_{\text{new}}) &= [0.340, 0.535], \\ P(y_{\text{new}} = 1 \mid \tilde{x}_{\text{new}}, \mathcal{D}) &= 0.436. \end{aligned}$$

The posterior predictive probability (0.436) is consistent with the observed proportion of successes in the dataset ($54/120 = 0.45$), suggesting that the model is a good fit.

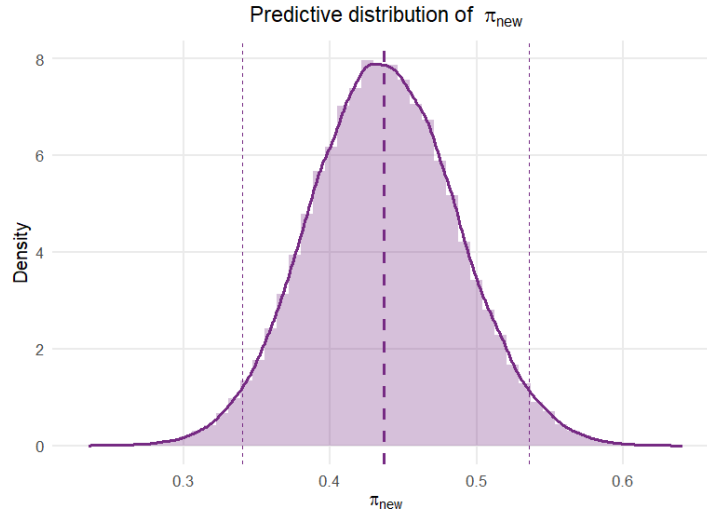


Figure 6: Posterior predictive distribution of $\pi_{\text{new}} = P(y_{\text{new}} = 1 \mid \tilde{x}_{\text{new}}, \beta_0, \beta_1)$ at $\tilde{x}_{\text{new}} = 0$. The dashed line marks the posterior mean; dotted lines mark the 95% credible interval.

5 Model Comparison

We now compares M1 against the model with only the intercept

$$M_0 : \quad \text{logit}(\pi_i) = \beta_0, \quad \beta_0 \sim \mathcal{N}(0, 3^2), \quad (5)$$

using the Deviance Information Criterion (DIC), available directly from the JAGS workflow via `dic.samples()`. The DIC decomposes as $\text{DIC} = \bar{D} + p_D$, where $\bar{D}$ is the posterior mean deviance and p_D is the effective number of parameters. Results are reported in Table 3.

Table 3: DIC comparison between M1 and M0.

Model	$\bar{D}$	p_D	DIC
M1 (full)	145.4	1.98	147.4
M0 (intercept)	166.1	0.99	167.1
$\Delta\text{DIC} = \text{DIC}(\text{M1}) - \text{DIC}(\text{M0})$			-19.7

M1 achieves a substantially lower DIC ($\Delta\text{DIC} = -19.7$). By standard guidelines, $|\Delta\text{DIC}| > 10$ is considered decisive evidence in favour of the model with the lower value. The effective number of parameters is $p_D \approx 2.0$ for M1 and $p_D \approx 1.0$ for M0, both consistent with the respective model structures. The improvement in $\bar{D}$ (-20.7) far outweighs the small increase in model complexity ($\Delta p_D = +0.99$), confirming that the inclusion of x substantially improves predictive fit without meaningful overfitting.

6 Discussion

All convergence diagnostics indicate reliable sampling. The Gelman–Rubin statistic is $\hat{R} = 1.00$ for all parameters, the effective sample sizes exceed 47 000, and autocorrelation is negligible beyond lag 5. The trace plots confirm that all four chains mix well and stabilise well within the burn-in period.

The posterior uncertainty for β_1 is moderate ($\text{SD} = 0.25$) but concentrated well above zero: the entire 95% credible interval excludes zero, making the evidence for a positive covariate effect credible and robust. The intercept β_0 carries more relative uncertainty and its credible interval includes zero, which is consistent with the near-balanced outcome rate at the mean of x .

The covariate x has a positive and credibly non-zero effect on the log-odds of success. The posterior odds-ratio $e^{\beta_1} \approx 2.77$ implies that a one-standard-deviation increase in x nearly triples the odds of a positive outcome, a practically meaningful effect size.

The posterior predictive probability at the mean of x is 0.436, very close to the observed success rate of 0.45. The 95% credible interval for π_{new} is well calibrated and contains the empirical proportion, supporting the adequacy of M1.

The DIC comparison decisively favours M1 over M0 ($\Delta\text{DIC} = -19.7$). The covariate x provides substantial additional predictive information beyond the intercept alone, and its inclusion is clearly supported by the data.